Teaching Statement

Jiashu Han

My experience as a teaching assistant since 2019 and as an instructor of record in summer 2026 has led me to believe that teaching is a collaborative and reflective process in which the instructor and the students share responsibility for meaningful learning. Students learn most effectively when they actively construct their own mental models, rather than passively receiving information. This principle matters even more now that AI can produce answers on demand. As an instructor, my role is to provide students with a conceptual foundation, help them connect ideas, and give them opportunities to test their reasoning.

I put these beliefs into practice when I designed and taught PHYS 1A (Mechanics) across both summer sessions in 2026 at UC San Diego, with more than 80 students in total. Most of the students were life-science majors taking the course to meet their major or pre-professional requirements. Being responsible for every part of a course, from the learning outcomes to the design of the course Canvas page, showed me where my philosophy held up and where it needed to grow.

Designing a student-centered course

Effective teaching begins with a backward design process, which establishes learning goals that fit both the curriculum and the students, before building assessments or lessons. Because PHYS 1A has a department-designated textbook and shared expectations, I began preparing three months early by meeting with past instructors to understand those expectations before writing my own outcomes.

Those conversations also taught me about the students I would be teaching: mostly life-science majors with minimal background in physics who needed physics as a foundation for their fields. Alongside the core topics, I therefore added fluid mechanics and chose examples that connect physics to their disciplines. The power of relevance became clear to me during the unit on fluid pressure. In one lecture, I asked students to determine how high a bag of IV solution must be hung for the fluid to flow into a vein. Several students told me how excited they were to see a medical application in mechanics, and that connecting it to their own experience helped the idea of fluid pressure stick. This experience led me to recognize relevance as one of the most effective forms of scaffolding I can offer.

To help students make sense of this material on a compressed schedule, I organized my lectures into modules that followed the same arc: introduce a concept, motivate it with a simple example, pause for an in-class poll and discussion so students can process the new information, show a live demonstration where possible, integrate the new concept with earlier material in a more challenging example, and then close with a summary of problem-solving tips. This predictable structure provided students with a consistent path from exposure to application, with regular moments for them to do the thinking themselves.

Lowering barriers and building trust

Equity is central to my teaching, and it means removing obstacles that have nothing to do with physics. A course with many kinds of assignments asks students to spend real effort on logistics: what is due, when, and how to balance it all. To scaffold the logistics as well as the content, I consolidated the course's Canvas site into a single page and laid out every task in a day-by-day calendar, so students could devote more attention to learning and less to tracking deadlines. Students noticed, and one called it the "most student-friendly Canvas interface I've ever seen."

Resources only help if students feel welcome to use them. I build relationships based on trust, care, and mutual respect, beginning with the language of my syllabus policies. I held office hours in the classroom immediately after lecture and added afternoon office hours for those who could not stay. In addition, I emailed students who struggled on assessments to offer help and ran a mid-course survey in each session to adjust the course. In both sessions, students rated the classroom as welcoming and supportive above the mean for other lower-division physics courses in the same session, and one wrote that this support "*made learning less intimidating.*"

Assessing for learning

Creating an environment where students feel safe asking questions is only the first step. Those questions also reveal where instructors and students see the material differently. I first became aware of this cognitive gap in my own teaching as a teaching assistant in 2019–21, when I often worked through problems at the board. In every lecture-course discussion section I taught during those years, my lowest-rated item concerned whether my explanations were pitched at the right level. Students described steps I skipped. Connections that seemed obvious to me were precisely the ones some students needed explained, yet they often struggled to put their questions into words. In a Winter 2020 lab, where I coached small groups at their tables, 91% of respondents agreed that my explanations were pitched at the right level. By 2026, I was giving students more time to work through problems in discussion and using weekly feedback to adjust practice problems. One student wrote, "*you evolved the formats to help us prepare for quizzes,*" and another, "*all students are properly challenged, but no one is left behind.*"

Frequent, low-stakes formative assessment is how I bridge that gap and ensure students do the cognitive work themselves rather than relying on AI tools for homework. In PHYS 1A, I layered in-class poll questions, short daily review quizzes, weekly homework, and weekly in-class quizzes. Because I had already built live polls into every lecture arc, I could use them to surface and address misconceptions in real time. After each weekly quiz, students could earn back points by correcting their mistakes, turning assessment into an opportunity to take ownership of their learning.

After a peer observed one of my lectures during the first summer session, I added prompts to my slides reminding me to pause, invite questions, and read the room, and adjusted practice problems to focus on choosing principles and setting up the solution. In end-of-term evaluations from both sessions, in-class practice was the most frequent theme in students' comments about what helped them learn; one described a cycle of "*in class questions to see where my knowledge started, then… daily review and homework to help solidify.*" Some still wanted more time to work problems themselves.

Balancing rigor and motivation

My most important lesson this summer concerned the relationship between rigor and motivation. I have always believed that learning outcomes should not be watered down to make a course easier to pass, and I designed my weekly quizzes to fully reflect the learning outcomes I established at the beginning of the course. Yet the quiz averages hovered around 50%, even though final grades were in line with expectations. In conversations after quizzes, some students could explain individual concepts but struggled to connect them in a multistep problem. Their scores gave little indication of which parts they already understood, and their morale suffered.

End-of-term evaluations showed that some students felt that the quizzes were a larger step beyond the homework and in-class practice than they had expected. Although the quizzes were cumulative, I had explained their scope only on page four of the syllabus. My quiz questions also targeted the course's most demanding end goals. Most of the questions asked students to interpret a physical scenario and then solve for a quantity, a task that layers conceptual and analytical skills at once. When I teach again, I will sort my learning outcomes by difficulty and include quiz questions at each level, ordered from easy to hard, so students can build confidence as they go. Their answers and scores would also show which skills they have developed and where they need more practice. In addition, I will state the quiz scope at the top of the syllabus and before each quiz, give a practice quiz in the same format, and share more detailed rubrics so students can see the reasoning behind their grades. Rigor remains the goal, but well-designed scaffolding is what lets students meet it with confidence.

Looking ahead

Teaching my own course clarified how I want to evolve my instruction — both to foster peer collaboration and to prepare students for a landscape where AI tools can solve standard textbook problems on demand.

First, in response to students' requests for more problem-solving time, I want collaboration to play a larger role in lectures. I plan to open with brief icebreakers and dedicate significantly more class time to think-pair-share and small-group problem solving, circulating among groups as I did in labs and turning to peer discussion whenever polls reveal widespread confusion.

Second, I want to make the reasoning habits emphasized in my lectures an explicit part of student practice. These include planning a setup and judging whether a solution is physically sound. Alongside frequent in-class assessment and worked examples that prioritize setup over arithmetic, I plan to give students flawed solutions, including AI-generated ones, and ask them to identify the first step the physics does not justify. This will extend the reflective work they already do when correcting their own quizzes.

Ultimately, designing and teaching PHYS 1A showed me that rigor and accessibility need not compete when instruction and assessments are scaffolded toward ambitious goals. My aim as an educator is to help every student build the conceptual confidence to construct, critique, and own their reasoning, and to leave my classroom prepared to engage deeply with complex ideas.